Evidence and Experimental Design for Sovereign Agent Systems

Abstract

A1 — A locally operated agent organization is a testable systems proposal; its end-to-end autonomy is Unproven. Confidence: high in this evidence classification. This critical review examines six components: autonomous orchestration, model routing, failure observability, durable recovery, organizational memory and training-data attribution. Primary literature supports several constituent mechanisms, including feedback-conditioned agent behavior, cost-sensitive routing, destination-side duplicate suppression, content-addressed storage and predictive retrieval caching. It does not establish their composition as reliable unattended operation. The principal research opportunity is a reproducible integration with explicit authority, recovery and evidence boundaries. Broad architectural novelty and training attribution from output alone are not supported. A local case study supplies useful implementation findings but no completed confirmatory experiment. The proposed research program therefore separates correctness, resource feasibility, causal performance improvement and provenance. **Evidence that changes this conclusion:** independently verified execution of the registered component studies and recovery controls, within a declared deployment envelope.

Scope and evidence method

0.1 — This is a critical integrative review, not an exhaustive systematic review or a completed experimental paper. Supported; confidence high. The evidence base comprises 27 public primary sources: research papers, original systems reports, official implementation documentation and a statistical reference. Targeted searches covered agent reflection and failure taxonomies; routing; durable execution, duplicate suppression and fencing; capability security; observability; memory hierarchy and retrieval caching; canary extraction; and multiagent debate. Citation tracing followed the proposal's references and the closest contrary mechanisms. Central methodological sections were inspected where available; a listed research paper is not automatically treated as peer reviewed. The source inventory records versions, locators, publication status and applicability limits. Literature inspection was principally completed on September 11, 2026; selected chronology and product metadata were rechecked on September 16 UTC. No search-completeness percentage, pooled effect size or publication-bias estimate is claimed. **Change criterion:** a reproducible broader search or inaccessible primary evidence can revise coverage and conclusions.

0.2 — Three evidence levels must remain separate. Supported methodological distinction; confidence high. “Supported” means supported within the stated source or observation scope; it does not certify the complete fleet. “Disputed” means a counterexample, incompatible evidence or an invalid inference defeats the stated formulation. “Unproven” means missing or inconclusive evidence. Confidence is an editorial assessment of the judgment, not an estimated probability of future success. Public literature, private case observations and proposed engineering choices are identified separately. Each numbered claim includes evidence that would change its verdict. Every proposed addition is marked **NEW SPECULATION**; the phrase identifies a design hypothesis, not an allegation that the underlying scientific principle is novel.

0.3 — The case study is dated and privately documented. Supported as an evidence description; confidence high. Local audit materials recorded on September 11 include source excerpts with hashes, a bounded host/corpus inventory, a frozen failure-ledger prefix and draft broker, assay and HOLD protocols. They are cited as **L1**, with original moonshot lines labeled **M** and the salon protocol labeled **S**. These materials are not peer-reviewed publications, proof of historical priority or evidence of the current running system. No completed HOLD-3/M or training-assay receipt was available. The proposed September 12 pilot review is now past; its unexecuted claims remain Unproven. September 20, 23:59:59 UTC remains a proposed review point, not a commitment to execute. **Change criterion:** new dated execution receipts, retained alongside the older record, can update those states without rewriting history.

1 Autonomous orchestration and inherited lessons

1.1 — Feedback-conditioned agent improvement has prior support; automatic beneficial evolution does not follow. Supported mechanism, Unproven fleet gain; confidence high. Reflexion stores linguistic feedback in memory and conditions subsequent attempts on it, without requiring weight updates. Its results and ablations support the usefulness of particular feedback and memory arrangements. Its limitations include local optima and difficulty constructing reliable tests for impure or concurrent programs. The fleet's folded “genetics” in M:32,41,67 belongs to this broad family; extracting a lesson is not itself evidence that the next agent improves. A lesson can preserve an erroneous diagnosis, overfit a task or introduce an unsafe instruction.[1] **Change criterion:** held-out, paired task outcomes show that the inherited content adds value beyond extra context, extra attempts and evaluator leakage.

1.2 — The self-correction literature is conditional and contested. Disputed as either a universal benefit or a universal impossibility; confidence high. Huang and colleagues find that tested intrinsic reasoning correction can fail or degrade accuracy without external feedback. Liu and colleagues report countervailing results with neutral prompts and zero-temperature decoding. The studies differ in conditions and should not be collapsed into a single universal rule. Neither establishes reliable long-horizon institutional learning across arbitrary tasks.[2, 3] **Change criterion:** a preregistered comparison using the fleet's actual models, tasks, prompt construction and feedback channels supports a bounded conclusion. A title-level citation is insufficient to settle the disagreement.

1.3 — Orchestration introduces its own failure surface. Supported concern, Unproven local rate; confidence high. The MAST study organizes multiagent failures into specification/system design, inter-agent coordination and verification/termination categories. This establishes reasons to inspect interaction traces rather than score model answers alone; its observed frequencies cannot be transferred directly to a different fleet.[4] A local two-failure rule can be bypassed if a fold, replacement process or fresh request identifier resets the counter. Spawn ceilings can likewise fail when checked independently by concurrent parents. These are state-management hypotheses, not demonstrated production incidents in this review. **Change criterion:** independent admission and receiver traces show persistent operation budgets, atomic global admission and monotone authority across replacement.

1.4 — A learning gate should separate learning value from permission to deploy. Unproven proposal; confidence medium. NEW SPECULATION. Retain three arms: frozen parent, candidate lessons and token-length-matched irrelevant lessons. Use held-out tasks, equal opportunity budgets and a frozen scorer; preserve failures and all fold/evaluation costs. Candidate prompts must pass a separate authority and injection gate before promotion. The inherited 768-episode plan is a candidate sample budget, not a literature-derived minimum or a power calculation. At the September 20 review, a paired

lower confidence bound exceeding the proposed five-point score margin would support the registered gain; an interval crossing the boundary remains Unproven. Any policy-expanding lesson blocks promotion regardless of mean performance. **Change criterion:** a pre-seal power analysis or disjoint pilot may justify a different design before outcomes are observed.

Source notes

1. Noah Shinn et al. (2023). Reflexion: Language Agents with Verbal Reinforcement Learning.

2. Jie Huang et al. (2024). Large Language Models Cannot Self-Correct Reasoning Yet.

3. Dancheng Liu et al. (2024). Large Language Models Have Intrinsic Self-Correction Ability.

4. Mert Cemri et al. (2025). Why Do Multi-Agent LLM Systems Fail?.

2 Routing resource allocation and authority

2.1 — Learned routing is established as a method, while fleet economics remain empirical. Supported method, Unproven transfer; confidence high. RouteLLM learns strong/weak model routing from preference information and evaluates cost-quality tradeoffs. Its limitations explicitly recognize shifts between benchmark and application distributions. A six-class registry and classifier, as proposed in M:43, therefore enter an existing research area; their value depends on local calibration, eligibility and accounting.[5] A rarely used specialist may be fully healthy. Conversely, recent usage does not prove readiness. Quota exhaustion, missing credentials, capability mismatch, cold-start delay and deliberate idleness should not share one STALLED label. **Change criterion:** a frozen in-domain workload shows lower total measured cost at a declared quality constraint, including router overhead and failed assignments.

2.2 — Resource sharing is a hierarchy of separate capacities. Supported distinction; confidence high. Official Ollama documentation distinguishes concurrent loaded models from parallel requests to one model; context length and request parallelism affect memory demand.[6] Shared model weights can reduce duplication while per-request state and memory bandwidth still limit service. In L1, the host exposed 63.951 GiB physical RAM, while the Docker VM exposed 31.319 GiB. A simultaneous 40 GiB resident in-cage worker set could not fit that observed VM before existing services. A 40 GiB ceiling alone does not imply that demand, however. Eight logical executors are not equivalent to eight model copies or eight simultaneous generations. **Change criterion:** a new frozen configuration and workload trace establish the permitted counts, peak physical/committed memory, paging and observer survival.

2.3 — Scoped API credentials alone do not establish child-authority containment. Supported principle, Unproven implementation; confidence high. Capsicum demonstrates kernel-enforced restriction of ambient authority and attenuation of object rights. Its FreeBSD implementation is not evidence of equivalent protection on Windows or WSL2. AgentDojo supplies a relevant adversarial evaluation pattern: untrusted material enters a tool-using agent's context while the agent retains legitimate work to perform.[7, 8] Applied to this fleet, a child must be unable to regain forbidden access through inherited handles, shell execution, alternate tools or an unmediated filesystem path. Prompt instructions expressing child≤parent are not the enforcement boundary. **Change criterion:** adversarial child processes and retrieved instructions fail to exceed receiver-enforced rights while authorized tasks remain usable.

2.4 — Routing and autonomy need a bounded economic experiment. Unproven proposal; confidence medium. NEW SPECULATION. Compare capable-static assignment, cheapest-static assignment and proposed routing on the same frozen requests. If cash cost is zero in all arms, a cash-saving percentage is undefined; use a declared alternative such as metered energy or occupied

inference time, while reporting subscription allocation, quota and operator minutes separately. The existing candidate gate—cost-reduction lower bound at least 20%, score-loss upper bound below two points—is a chosen engineering margin. Record queue age and task acceptance, so cheap indefinite waiting cannot win. At September 20, judge each registered endpoint from complete attempt and cost ledgers. **Change criterion:** pre-seal workload evidence can alter the margin or cost unit; post-result convenience cannot.

Source notes

5. Isaac Ong et al. (2025). RouteLLM: Learning to Route LLMs with Preference Data.

6. Ollama contributors (n.d.). FAQ — How does Ollama handle concurrent requests?.

7. Robert N. M. Watson et al. (2010). Capsicum: practical capabilities for UNIX.

8. Edoardo Debenedetti et al. (2024). AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents.

3 Failure observability and measurement validity

3.1 — Typed return rows are necessary accounting aids but incomplete outcome evidence. Supported distinction; confidence high. Google's SRE guidance separates internal instrumentation from externally observed behavior and explicitly recognizes wrong content returned with HTTP 200 as an error. It also emphasizes latency, traffic, errors and saturation, including tail behavior.[9] A semantic health endpoint therefore needs task-specific postconditions and a missing-data policy. Its own process can fail while a healthy-looking last value persists. The proposed independent observer should reconcile admitted intents, observed dispatches, receiver effects and terminal dispositions without relying exclusively on the emitter it audits. **Change criterion:** wrong-artifact, missing-row, duplicate-row and observer-loss injections are detected from independent evidence within the declared deadlines.

3.2 — The local failure count illustrates a measurement-grain error. Supported local observation; confidence high. L1's frozen 27,934,077-byte ledger prefix contains 24,854 records whose JSON signature exactly equals `run_shell:c73866f2f1b5`. Another 153 records mention that signature elsewhere, yielding 25,007 matching lines. The matched records lack explicit attempt identifiers; neither matching lines nor unique record hashes prove distinct executions. Ten malformed lines remain separately reported. The earlier repair receipt refers to a different byte count, so it cannot silently substitute for this snapshot. These findings support the existence of a persistent diagnostic signature, not a measured count of exactly 25,007 executions. **Change criterion:** an immutable attempt-linked query reconciles dispatch identity and preserves exclusions. The frozen query and digest belong with the claim.

3.3 — The appropriate middle-test endpoint is safe disposition rather than compulsory repair. Unproven proposal; confidence high in relevance. NEW SPECULATION. Inject a tool success envelope with a wrong artifact, suppress its completion event and duplicate an unrelated event while other tools succeed. A passing observer must detect both loss and duplication; net-zero count error is not reconciliation. A correct response may be APPLIED, final nonapplication or an explicit UNKNOWN hold. The existing five-second detection, 30-second UNKNOWN and 60-second post-fault reconciliation thresholds are fixture acceptance choices, not universal service requirements. At September 20, compare protected monotonic-clock records with exact injection times and deadlines. **Change criterion:** a realistic workload or fault model requires different thresholds before seal; false completion or an unsafe redispatch disputes the tested build immediately.

Source notes

9. Rob Ewaschuk (2016). Monitoring Distributed Systems.

4 Durable execution and effect reconciliation

4.1 — Cryptographic history and durable execution solve different problems. Disputed equivalence; confidence high. Temporal's architecture separates deterministic workflow decisions from side-effecting activities and requires activities to be idempotent or non-retryable. Replay works within that execution protocol; append-only history alone is not the protocol.[10] Hash linkage can make alterations detectable relative to a trusted commitment. It cannot reveal an effect whose acknowledgment was lost, recover absent object bytes or decide whether a delayed writer will still commit. The retracted “90% of Temporal” wording in M:13 had no defensible quantitative denominator. **Change criterion:** specify the event model, nondeterministic inputs, code-version compatibility and receiver guarantees, then demonstrate the declared subset under faults.

4.2 — Exactly-once effects are conditionally achievable, not supplied by a caller-side receipt. Supported; confidence high. RIFL couples durable completion records with the operation's effects and makes retries meet those records. It also addresses record reclamation and returning clients. This refutes both the overclaim that a local replay loop automatically produces exactly-once effects and the opposite overclaim that exactly-once semantics are universally impossible. Its assumptions concern cooperating infrastructure; its performance cannot be transferred to arbitrary HTTP tools.[11] In L1, /run_task dispatches directly to tools, while the named broker saves its receipt after effect execution. The integration gap survives a correct-looking broker module elsewhere. **Change criterion:** all admitted mutation paths reach a cooperating receiver that commits effect and dedup state atomically.

4.3 — A dead-worker judgment does not fence its outstanding writes. Supported; confidence high. Chubby's sequencers address delayed requests from former lock holders by making recipients check lock generations. Lock-delay fallback is not a complete equivalent.[12] A lease timeout, heartbeat or stale lock-file age therefore cannot by itself authorize an overlapping new writer. The same distinction applies on one machine whenever asynchronous requests, suspended processes or open file handles outlive the controller's judgment. **Change criterion:** suspend owner A, authorize and commit owner B, then resume A; the receiving resource rejects A's mutation. The authority assigning generations and the storage retaining them must also be specified.

4.4 — A transactional store closes only the transaction it actually owns. Supported; confidence high. SQLite's atomic-commit account depends on explicit storage, locking and flush assumptions. Its multi-file discussion concerns participating database files, not arbitrary filesystem mutations or remote effects; the cited account is principally the rollback-journal design, not a blanket guarantee for all WAL configurations.[13] A database row recording a separately executed file write leaves an ambiguity gap. The proposed local contract can instead make logical documents and ledger events canonical in one store and materialize plain files separately. That changes authority semantics and must be declared, not hidden behind the word “receipt.” **Change criterion:** either the effect participates in the same transaction or a separately tested receiving-resource protocol supplies equivalent invariants.

4.5 — UNKNOWN requires durable finality rules and a restoration horizon. Unproven implementation proposal; confidence high. NEW SPECULATION. Persist one immutable intent, a stable admission identity, separate attempt identities and a receiver-checked epoch. Reusing an operation ID with different intent is a conflict. A status query must return APPLIED with operation-bound evidence, NOT_APPLIED_FINAL with an effective cancellation/fence barrier, or UNKNOWN. Mere absence or matching desired bytes without provenance is insufficient. Retain deduplication knowledge through the replay/backup horizon, or reject expired client generations after reclamation; restoring an old controller must not resurrect a completed effect. At September 20, fault

tests must cover a delayed original request, an old backup and expired dedup state. **Change criterion:** a simpler protocol proves the same invariants for the declared destination set.

4.6 — A small executable state model can expose design errors before expensive agent trials. Unproven proposal; confidence medium. NEW SPECULATION. AWS's formal-methods experience reports subtle bugs found by model checking, while distinguishing design models from implementations and noting a missed liveness bug when liveness was not checked.[14] Model admission, reservation, ownership, dispatch, commitment, dedup expiry, restore and UNKNOWN transitions. Check both safety—no duplicate or unauthorized commitment—and liveness under declared recovery assumptions. Calibrate the model with a deliberately broken protocol, then test whether implementation traces conform. A finite successful model neither proves unmodeled code correct nor predicts sustained latency. **Change criterion:** bounded models and fault traces agree on the registered cases by September 20; newly found counterexamples revise the model and implementation together.

Source notes

10. Temporal contributors (n.d.). Temporal system architecture.

11. Collin Lee et al. (2015). Implementing Linearizability at Large Scale and Low Latency.

12. Mike Burrows (2006). The Chubby lock service for loosely-coupled distributed systems.

13. SQLite project (n.d.). Atomic Commit In SQLite.

14. Chris Newcombe et al. (2015). How Amazon Web Services Uses Formal Methods.

5 Organizational memory and architectural novelty

5.1 — Shared memory-hierarchy ideas do not establish architectural identity with Engram. Disputed identity claim; confidence high. Engram maps token n-grams into learned tables and fuses retrieved vectors with contextual hidden states. Its deterministic keys enable advance host-memory fetching. A corpus service instead maps document or chunk identities into external representations that an agent retrieves. The key domain, learned state, update semantics and fusion mechanism differ. Engram v1 was published January 12, 2026, before the reported July–August private fleet records; September product shipment is not the earliest public comparison date.[15] Chronology alone neither proves copying nor disproves independent development. **Change criterion:** authenticated earlier artifacts change chronology; an explicit mechanism mapping changes the architectural comparison. Local timestamps alone do not settle priority.

5.2 — External knowledge caching and predictive personal-memory caching have substantial prior art. Disputed broad novelty; confidence high. RAGCache caches intermediate states of retrieved documents, accounting for their prefix context across GPU and host memory. TeleRAG predicts vector clusters needed at subsequent RAG stages and overlaps their movement with generation. PerCache predicts future queries from personal knowledge and query history to populate a hierarchy of cached intermediates and answers.[16, 17, 18] These are external-information mechanisms, not merely offloaded model weights. Their benchmark improvements are workload-specific. The PerCache metadata presents a submission-date/identifier discrepancy; this review relies on the inspected mechanism and does not claim an exact first-publication day. **Change criterion:** a direct feature comparison establishes a narrower residual contribution; a search with no matching title is not enough.

5.3 — Content identity does not supply semantic retrieval, freshness or authority. Supported distinction; confidence high. Venti is early prior for immutable content-addressed blocks, duplicate coalescing and reconstruction of a lookup index from stored data. Its historical hash choice is not a current security recommendation. Zanzibar separately addresses consistency between permission

changes and content access.[19, 20] The fleet's chain-leaf hash identifies the serialized record unless an explicit mapping binds it to chunks. Expected constant-time lookup assumes the key is already known; natural-language key discovery, similarity search, transfer and generation remain. A valid hash proves neither that the content is current nor that a revoked agent may read it. **Change criterion:** an explicit query model and version/authorization invariant cover every direct, semantic and derived-cache path.

Mechanism	Addressed object	Missing guarantee if used alone
Engram lookup	Learned n-gram parameters	Document provenance or user authorization
Content-addressed storage	Immutable bytes	Current truth or semantic relevance
Embedding index	Encoded chunks	Complete corpus coverage or freshness
Retrieved-prefix KV cache	Model state for a prefix	Validity under changed preceding context
Predictive prefetch	Expected future accesses	Positive net benefit under workload shift
Answer cache	Previously produced responses	Current evidence and permission validity

5.4 — The observed corpus path does not substantiate the proposed whole-corpus table. Unproven scale claim; confidence high in the gap. M:33,45 and L1 distinguish design arithmetic from implementation. A live metadata scan found 882,366 ordinary files while skipping a reparse point and encountering 62 unreadable directories. It neither verifies nor refutes a historical 1.3M-file denominator. An independent reproduction of the inspected text-cache selection found 328 readable files, 328 distinct snippet hashes and no embedding allocation in that path. The joined text was 378,181 UTF-8 bytes, not measured heap or peak residency. **Change criterion:** sealed eligibility, chunk-occurrence, unique-chunk and successful-slot manifests reconcile without unexplained omissions, followed by actual allocation and workload traces.

5.5 — The memory estimate is conditional and must include construction peaks. Supported arithmetic, Unproven deployment; confidence high. Nineteen million 256-dimensional fp16 vectors require 9,728,000,000 bytes. Binary 32-byte keys add 608,000,000 and uint64 slot IDs add 152,000,000: 10,488,000,000 bytes, or 9.768 GiB, before index slack, text, metadata, runtime objects and working copies. Another full vector copy adds 9.060 GiB. DeepSeek's V4.1 report separately states 8B active parameters in prefill, 16B in decode and 890 bytes/token of global KV; none is a total-memory figure.[21] Sparse activation does not remove the need to store or fetch inactive weights. **Change criterion:** peak measurements on the actual model, KV representation and cache build replace estimates. A compression factor from another architecture cannot fill an unmeasured budget cell.

5.6 — The publishable residual is a systems integration hypothesis with comparative tests. Unproven; confidence medium. NEW SPECULATION. Define the candidate contribution as permission-aware, content-addressed organizational memory with replay-linked provenance and predictive fetching under changing workloads. Compare uncached, reactive-cache, proposed-predictor and oracle-prefetch arms. The oracle is a headroom bound, not a deployable competitor. Freeze model, tokenizer, context, KV format and resource budget; use chronological holdouts and cold, warm and shifted traces. Meter search, fetching, prediction, wasted transfers, prompt assembly, prefill and decode separately. The inherited ≥10% p95 improvement and <2-point score-loss margins are proposed acceptance rules. Any unauthorized or superseded content served as current kills the corresponding

invariant. **Change criterion:** September 20 traces demonstrate benefit after all costs; a distinct memory/no-memory study establishes task usefulness. Neither experiment establishes absence of prior art.

Source notes

15. Xin Cheng et al. (2026). Conditional Memory via Scalable Lookup: A New Axis of Sparsity for Large Language Models.

16. Chao Jin et al. (2024). RAGCache: Efficient Knowledge Caching for Retrieval-Augmented Generation.

17. Chien-Yu Lin et al. (2026). TeleRAG: Efficient Retrieval-Augmented Generation Inference with Lookahead Retrieval.

18. Kaiwei Liu et al. (2026). PerCache: Predictive Hierarchical Cache for RAG Applications on Mobile Devices.

19. Sean Quinlan, Sean Dorward (2002). Venti: A New Approach to Archival Data Storage.

20. Ruoming Pang et al. (2019). Zanzibar: Google's Consistent, Global Authorization System.

21. DeepSeek-AI (2026). DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression.

6 Canary attribution and controlled extraction

6.1 — Reproduction is not, by itself, a training-mechanism identifier. Supported distinction; confidence high. Zhang and colleagues analyze why ordinary membership inference cannot generally establish sound training-data proofs for opaque production models, while identifying random canaries, watermarking and extraction as conditional alternatives. Their account also distinguishes controlled laboratory evaluation, where training inclusion is known, from opaque-model attribution.[22] A tool-enabled service can repeat a secret from retrieval, conversation history, caches or another component without its model weights having learned it. The candidate secret's independence from the tested output mechanism is therefore load-bearing. **Change criterion:** authenticated inclusion and exclusion of competing routes support a scoped training interpretation. A hash registry establishes commitment to bytes, not the route by which another system obtained them.

6.2 — Ten duplicates is not a physical threshold for memorization. Disputed universal threshold; confidence high. Carlini and colleagues observe dependence on duplication and prefix length, including memorization with few duplicates. Extraction can change when the supplied prefix changes even though weights remain fixed.[23] A negative result after a chosen dose thus combines training sensitivity, query sensitivity and model behavior; it does not demonstrate that training did not occur. **Change criterion:** a controlled dose–response study with authenticated inclusion estimates detectability for a specified recipe and query set. It cannot turn that estimate into a universal minimum or a vendor-wide negative inference.

6.3 — Four exposure arms can test causal discrimination only with controlled custody. Unproven protocol; confidence medium. NEW SPECULATION. Generate sixteen independent uniform 128-bit suffixes, four each for neither, retrieval-only, history-only and authenticated known-training. The unexposed base must predate suffix generation. Retrieval and history arms receive exactly one registered record through their designated route. The training arm receives a frozen base plus authenticated adapter and no suffix-bearing inference context. Two fixed suffix-free prompts per marker and fresh inference state constrain the query family. A generic decoder permits exactly 32 lowercase hexadecimal characters without enumerating the correct secrets. This deliberately measures constrained extraction, not spontaneous ordinary prose.

The custodian protects registry, mappings, training records and raw I/O. Commit the recipe, seeds, schedule and salted truth bundle before training; commit effective checkpoint hashes after training and before scored inference; freeze independently written verifier programs before outcomes. Both workers receive sanitized match receipts and commit verdicts before provenance reveal. Before reveal, outputs

alone leave training Unproven even in the actually trained arm. After reveal, training receipts establish exposure and outputs measure extraction. **Change criterion:** custody attacks, route audits and disjoint positive controls pass before the September 20 confirmatory run. A trusted local administrator remains inside the threat boundary.

6.4 — The query cap must count every tested candidate, including windows and versions. Supported arithmetic, Unproven enforcement; confidence high. NEW SPECULATION schedule. The candidate schedule contains 32 primary outputs, 400 sensitivity outputs and 80 sham outputs: 512 in total. Comparing each with all sixteen suffixes reserves 8,192 comparisons. Sixteen registered retrieval/history input exposures add 256, for 8,448 of a 10,000 cap. The unused 1,552 are not permission for adaptive searching. A 64-character hex output contains 33 possible 32-character windows; extra generations or searched versions are charged even when they violate the protocol. Rechecking the same frozen bytes is reproducibility rather than another generated candidate.

For a candidate independent of a uniform 128-bit secret, the probability of a match is at most 2^{-128}; a union bound over Q such comparisons is at most $Q/2^{128}$. At Q=10,000 this is about 2.94×10^{-35}. This calculation is not a contamination bound, a posterior probability of training or a bound for deliberately exposed comparisons. **Change criterion:** two independent parsers reproduce the complete candidate ledger and identify null-eligible comparisons. Missing generated outputs leave the claimed bound Unproven at September 20.

6.5 — Sensitivity requires replication at the training-run level. Unproven proposed study; confidence medium in design and low in forecast extraction. NEW SPECULATION. Use ten training seeds at each of five doses—1, 3, 10, 30 and 100 optimizer-consumed unmasked occurrences per marker—plus ten zero-dose shams. Sixty fresh adapters share one fixed base and four training markers; this estimates seed variability conditional on that panel, not arbitrary-model or arbitrary-marker generalization. LoRA is a candidate way to reduce adaptation cost, not evidence that this suffix recipe will succeed.[24] The registered endpoint is whether a checkpoint reproduces any designated training marker under either fixed prompt. Keep missing checkpoints visible; do not count every prompt as an independent training replication.

Use six simultaneous two-sided exact binomial intervals, allocating alpha=0.05/6 to each. The binomial interval method is appropriate to the declared Bernoulli endpoint when its sampling assumptions hold.[25] With ten successes, the lower bound is $(0.05/12)^{(1/10)}\approx0.578$; with zero successes, the upper bound is approximately 0.422. Thus ten clean shams do not demonstrate a very low empirical contamination rate. The proposed high-dose lower-bound target >0.50, no sham hit and valid controls is deliberately narrow. **Change criterion:** a disjoint feasibility/power study can change sample size before seal; September 20 results must retain the chosen complete family, missingness and intervals without post-result extension.

6.6 — A positive assay does not restore production salting as attribution armor. Supported scope limit; confidence high. The strongest permissible result is that this pinned base, adapter recipe, fixed marker panel, dose and prompt family exhibited measured extraction after authenticated training, with alternative inference routes excluded. Retrieval/history positives demonstrate the ambiguity of output-only interpretation. The experiment does not establish remote-provider ingestion, ingestion time, authorship, ownership or historical priority. **Change criterion:** a separately registered deployment study supplies the additional provenance required for a broader claim. Until then, production salting remains outside the supported conclusion, whatever the local extraction rate.

Source notes

22. Jie Zhang, Debeshee Das, Gautam Kamath and Florian Tramèr (2025). Position: Membership Inference Attacks Cannot Prove that a Model Was Trained On Your Data.

23. Nicholas Carlini et al. (2023). Quantifying Memorization Across Neural Language Models.

24. Edward J. Hu et al. (2021). LoRA: Low-Rank Adaptation of Large Language Models.

25. NIST/SEMATECH (n.d.). Confidence intervals for a binomial proportion.

7 Independent verification and the research program

7.1 — Independence should be measured against access and truth, not compulsory disagreement. Disputed collapse rule; confidence high. Wynn and colleagues report harmful correctness changes after peer reasoning; Yao and colleagues study sycophantic convergence within particular debate configurations. Both report scope-dependent behavior. Neither supplies the rule in S:27 that isolated first-pass agreement itself proves collapse.[26, 27] Two workers can agree because evidence is decisive, because they share a flaw or because independence failed. Agreement alone does not distinguish those cases. **Change criterion:** calibrated truth cases and pre/post-interaction accuracy establish whether a proposed collapse detector predicts error. A known-correct unanimous case must not be rejected merely for unanimity.

7.2 — A two-worker HOLD should test each verifier against truth before comparing workers. Unproven protocol; confidence high in the distinction. NEW SPECULATION. Freeze inputs and hidden fixture truth; then freeze each independently implemented verifier; then issue the final seal. Each worker receives the same read-only inputs but a separate writable clone, with denied access to its peer and the hidden oracle. A may reconstruct state from events while B independently joins receiver facts to admitted identities. Both must reject intentionally broken deduplication, wrong artifacts, missing events, stale owners and leaked answers. Commit salted verdicts before reveal. Judge each against truth first and compare them second. Identical integer results are acceptable; identical mistakes are failure. **Change criterion:** actual access-control and calibration receipts establish this procedural independence by September 20. Shared-host compromise remains outside the claim.

7.3 — The first pilot should isolate the highest-value ambiguity boundary. Unproven proposal; confidence high in relevance. NEW SPECULATION. One deterministic worker admits one ledger insert. The receiver atomically commits effect and dedup; a protected controller kills the worker before its response; restart reuses the operation ID. Require one admission, at least two delivery attempts and exactly one canonical effect, supported by a destination oracle and independent clock. Keep known-bad controls outside the one scored case but in the cost record. No model is necessary. Proposed caps are 2 GiB worker and 1 GiB controller/observer, with 4 GiB measured spare headroom. **Change criterion:** a newly dated pilot seal and execution receipt establish this adapter/build/process-crash path only. The missed September 12 review does not make this a completed result.

7.4 — Full scale is a separate registration decision. Unproven; confidence medium. NEW SPECULATION. The inherited HOLD schedule contains 16 scenario cases per worker, 32 total, plus concurrency 1/2/8 × five crash boundaries ×100 fault targets per worker: 3,000 targets overall. Because each target has c−1 background operations, the matrix contains 11,000 logical operations before scenario operations and retries. Clones run sequentially so eight is a host-wide ceiling. If preflight admits only concurrency 1/2, register a smaller profile and keep eight-way feasibility Unproven. Shared implementation and schedules invalidate an interpretation of 3,000 trials as 3,000 independent

deployments. **Change criterion:** September 20 receipts reconcile enrolled, completed, aborted and missing cases against the admitted profile; any hard-invariant counterexample blocks that build.

7.5 — Artifact closure should precede expensive trials. Unproven readiness; confidence high. NEW SPECULATION. The required build order is scope/claim/supersession manifests; host and resource preflight; active-route and authority manifests; immutable intent and canonical-encoding rules; receiver and reconciler; frozen chain plus object-store closure; meter/observer implementation; protected fixture controller and truth; verifier A; independent verifier B; calibration/isolation/restore receipts; and final signed dependency closure. Fleet implementers build runtime components, two workers own their distinct verifiers, and the operator protects truth and signs. A specification supplies acceptance rules but is not an executable artifact. Every meter needs a formula, unit, denominator, raw source, trust boundary, sampling rule, threshold, missing-data treatment, implementation digest and known calibration result. **Change criterion:** the complete dependency manifest and accepted receipts exist by the declared seal date. Any missing mandatory field blocks that scope.

Source notes

26. Andrea Wynn, Harsh Satija and Gillian Hadfield (2025). Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate.

27. Binwei Yao et al. (2025). Peacemaker or Troublemaker: How Sycophancy Shapes Multi-Agent Debate.

8 Interpretation limitations and publication criteria

8.1 — Correctness and performance require different evidential standards. Supported methodological distinction; confidence high. One valid duplicate effect can refute a no-duplicate invariant, whereas a null performance difference may simply be inconclusive. The confirmatory performance studies need frozen primary endpoints, sample-size justification and uncertainty intervals appropriate to their dependence structure. Paired tasks should remain paired; temporally correlated cache traces need blocks or another explicit dependence model. Repeated requests sharing a cache or trained checkpoint are not automatically independent observations. Counts, uncertainty and all enrolled failures should be published; missing information should not be imputed as success. **Change criterion:** explicit sampling assumptions and sensitivity analyses justify a simpler estimator. These requirements apply at September 20 to the proposed studies, not retrospectively to exploratory results.

8.2 — The principal local findings are informative but narrower than a system evaluation. Supported limitation; confidence high. L1 observed a constrained VM, a bounded text-cache path, a dispatch/broker integration gap and a failure-count grain mismatch. It did not execute the eight-way workload, build the whole-corpus vector table or train the assay models. Source correctness also differs from activation: the shell-repair receipt reports old code still loaded in some processes. An initial audit inference that Windows signal zero necessarily terminates a process was corrected after a disposable-child calibration and recognition of the CTRL_C_EVENT branch. Retaining that correction is part of the evidence record. **Change criterion:** independent replication and fresh runtime receipts expand or contradict these observations. A dated snapshot must not be presented as current fleet health.

8.3 — The most defensible publication is a bounded systems contribution with open failure evidence. Unproven prospective contribution; confidence medium. NEW SPECULATION. A manuscript could contribute a specified local authority/effect protocol, a versioned organizational-memory implementation, and reproducible evaluation across crash, revocation, workload-shift and operator-absence conditions. Novelty must be argued by mechanism against prior systems; usefulness by comparative experiments; safety by invariants within a declared fault model; and provenance by custody evidence. Report model versions, machine configuration, workload eligibility,

prompts, failed runs and operator labor. Remote subscriptions may remain optional resources, but an offline claim needs a frozen set of tasks that complete without them. **Change criterion:** artifacts and independently reproducible results substantiate those contributions. Before that point, the appropriate publication category is a design/research protocol, not a validated autonomous-system result.

8.4 — The remaining highest-leverage enhancement is a release-activation and supersession gate. Unproven; confidence medium. NEW SPECULATION. Bind each runnable route to its actually loaded executable, runtime, adapter, policy and current instruction projection, then require a behavioral probe before admission. Keep desired and observed versions distinct. This addresses a concrete composition risk: corrected files, contracts and citations can coexist with old live workers and old prompts. Preserve history while preventing superseded rules from regaining authority. At September 20, the denominator is every enrolled runnable route/process instance; stale build, stale policy, superseded prompt and bypass-path calibrations must all be rejected. **Change criterion:** a simpler enforceable mechanism achieves that complete coverage and remains live during recovery.

8.5 — The single protocol rule to cut is automatic collapse attribution from first-pass agreement. Disputed; confidence high. Keep isolated first passes, evidence commitments and correctness checks; remove the demand to manufacture disagreement. This conclusion follows from the identification problem in claim 7.1, not from an assumption that consensus is reliable. **Change criterion:** a validated detector demonstrates that agreement alone distinguishes error from correct independent convergence in the relevant setting. Without that evidence, agreement is an observation requiring explanation, not a verdict.

Local evidence and reproducibility record

L1 — Private case dossier, not external validation. Supported description; confidence high. The September 11 dossier comprises moonshot-v1-source.md (M:13,27–34,41–46,54,66–70); moonshot-v1-1-source.md (accepted-disposition annex); salon-scorecard-source.md (S:27); v2-inventory.md; v2-inventory-metadata.json; current-cache-reconciliation.json; effect-broker-evidence-v0.md; failure-ledger-recount.json; shell-fix-receipt-source.md; windows-liveness-calibration.json; and the broker, assay and HOLD registration drafts. Repository paths and source hashes reside in that dossier. The failure-prefix SHA-256 is 9ab1b9dff8b2001c2a82a7c092d2c63613daa744507df6ea2b551bbad1957b74. The archive and replay query permit scoped recounting; no public access or third-party reproduction is implied. **Change criterion:** a released reproducibility package, independently verified with appropriate data access, upgrades the evidence from private observation to replicated case evidence.

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